# MATH MAGIC

**Smart Tricks for Faster Calculations**

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### THE IDEA

Most calculations do not need more steps.

They need a better approach.

This guide brings together practical shortcuts, useful patterns, and simple mental methods to help you calculate faster.

# SQUARE & SQUARE ROOT TRICKS

## Numbers Ending in 5

When a number ends in **5**, you can square it very quickly.

**Step 1:** Look at the number before **5**.

**Step 2:** Multiply that number by the **next whole number**.

**Step 3:** Write **25** at the end of your answer.

**Example: $35^2$**

The number before 5 is **3**.

The next whole number after 3 is **4**.

Now multiply:

**3 × 4 = 12**

Finally, attach **25** to 12:

**1225**

So,

**$35^2 = 1225$**

**Example: $85^2$**

The number before 5 is **8**.

The next whole number after 8 is **9**.

Now multiply:

**8 × 9 = 72**

Attach **25** to 72:

**7225**

So,

**$85^2 = 7225$**

## Quick Rule

For any number ending in **5**:

**Number before 5 × next number + 25**

For example:

**$45^2$** → 4 × 5 = 20 → **2025**

**$65^2$** → 6 × 7 = 42 → **4225**

**$95^2$** → 9 × 10 = 90 → **9025**

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## Numbers Near 100

When a number is close to 100, start with **$100^2 = 10{,}000$** and adjust for the difference.

**$98^2$**

98 is **2 less than 100**.

$100^2 = 10{,}000$
Subtract 2 × 100 = 200
Add $2^2 = 4$

So:

**$98^2 = 10{,}000 - 200 + 4 = 9{,}604$**

**$103^2$**

103 is **3 more than 100**.

$100^2 = 10{,}000$
Add 3 × 100 × 2 = 600
Add $3^2 = 9$

So:

**$103^2 = 10{,}000 + 600 + 9 = 10{,}609$**

## Quick Rule

For numbers close to 100:

**Start with 10,000, adjust for the difference, then add the difference squared.**

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## Numbers Near 50

When a number is close to 50, start with **$50^2 = 2500$** and adjust for how far the number is from 50.

**$47^2$**

47 is **3 less than 50**.

$50^2 = 2500$
Subtract $2 \times 50 \times 3 = 300$
Add $3^2 = 9$

So:

**$47^2 = 2500 - 300 + 9 = 2209$**

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## Difference of Squares

This trick works when two numbers are the **same distance from a middle number**.

For example, **49 and 51** are both **1 away from 50**:

**49 ← 1 → 50 ← 1 → 51**

So instead of multiplying 49 × 51 directly, use:

**(Middle Number)² − (Distance)²**

For **49 × 51**:

The middle number is **50** and the distance is **1**.

**49 × 51**
**= $50^2 - 1^2$**
**= 2500 − 1**
**= 2499**

## Quick Rule

When two numbers are equally far from a middle number:

**Multiply the middle number by itself, then subtract the distance squared.**

**49 × 51 → $50^2 - 1^2 = 2499$**

**98 × 102 → $100^2 - 2^2 = 9996$**

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## Build the Next Square

Once you know one square, you can quickly find the square of the next number.

To get the next square:

**Add twice the current number, then add 1.**

**Example: Find $41^2$**

We know:

$40^2 = 1600$

Now:

$2 \times 40 = 80$

Add 1:

$1600 + 80 + 1 =$ **1681**

So:

**$41^2 = 1681$**

## Quick Rule

**Next square = Current square + twice the current number + 1**

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## Quick Square Check

A perfect square can never end in:

**2, 3, 7, or 8**

So:

**1538 → Not a perfect square**

This is a quick elimination trick, not a complete test.

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## Estimate a Square Root

You can quickly estimate a square root by finding the two nearby perfect squares.

**Example: $\sqrt{150}$**

We know:

$12^2 = 144$
$13^2 = 169$

So:

**$12 < \sqrt{150} < 13$**

150 is only 6 more than 144, so $\sqrt{150}$ is a little more than 12.

A good estimate is:

**$\sqrt{150} \approx 12.25$**

This is an estimate, not the exact answer.

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# CUBE & CUBE ROOT TRICKS

## Cube Near 10

When a number is close to 10, you can use 10 as the starting point.

**Example: $9^3$**

9 is **1 less than 10**.

Using the cube pattern:

$10^3 - 3(10^2)(1) + 3(10)(1^2) - 1^3$

$= 1000 - 300 + 30 - 1$

**$= 729$**

So:

**$9^3 = 729$**

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## Cube Near 100

The same idea works for numbers close to 100.

**Example: $101^3$**

101 is **1 more than 100**.

$100^3 = 1{,}000{,}000$

Add:

$3 \times 100^2 \times 1 = 30{,}000$
$3 \times 100 \times 1^2 = 300$
$1^3 = 1$

So:

1,000,000 + 30,000 + 300 + 1

= **1,030,301**

Therefore:

**$101^3 = 1{,}030{,}301$**

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## Find a Cube Root

The last digit of a cube can tell you the last digit of its cube root.

For example:

**2197**

The number ends in **7**.

From the cube pattern, a cube ending in 7 comes from a number ending in **3**.

So the answer must end in 3.

Now check:

**$13^3 = 2197$**

Therefore:

**$\sqrt[3]{2197} = 13$**

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## Another Cube Root Shortcut

**$\sqrt[3]{13824}$**

The number ends in **4**.

A cube ending in 4 comes from a number ending in **4**.

Now check:

**$24^3 = 13{,}824$**

Therefore:

**$\sqrt[3]{13{,}824} = 24$**

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### Common Cubes

$1^3 = 1$
$2^3 = 8$
$3^3 = 27$
$4^3 = 64$
$5^3 = 125$
$6^3 = 216$
$7^3 = 343$
$8^3 = 512$
$9^3 = 729$
$10^3 = 1000$

Knowing these makes many cube-root questions a recognition problem.

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# TABLE TRICKS

### Split the Digits Trick

For a larger number, split it into its individual digits and use those digits to help rebuild the multiplication table.

For example, split **41** into **4 | 1**.

Now multiply each digit by **1 to 10**:

| | **4** | **1** | **Result** |
|---|---|---|---|
| ×1 | 4 × 1 = 4 | 1 × 1 = 1 | **41** |
| ×2 | 4 × 2 = 8 | 1 × 2 = 2 | **82** |
| ×3 | 4 × 3 = 12 | 1 × 3 = 3 | **123** |
| ×4 | 4 × 4 = 16 | 1 × 4 = 4 | **164** |
| ×5 | 4 × 5 = 20 | 1 × 5 = 5 | **205** |

| | | | |
|---|---|---|---|
| ×6 | 4 × 6 = 24 | 1 × 6 = 6 | **246** |
| ×7 | 4 × 7 = 28 | 1 × 7 = 7 | **287** |
| ×8 | 4 × 8 = 32 | 1 × 8 = 8 | **328** |
| ×9 | 4 × 9 = 36 | 1 × 9 = 9 | **369** |
| ×10 | 4 × 10 = 40 | 1 × 10 = 10 | **410** |

## How to Use It

Take **41 × 7** as an example.

Split **41** into **4 | 1**.

Then:

**4 × 7 = 28**
**1 × 7 = 7**

Using the place values and carry:

**28 | 7 → 287**

So:

**41 × 7 = 287**

The same process can be used to build the complete table from **×1 to ×10**.

## Quick Memory Pattern

Instead of trying to memorize:

**41, 82, 123, 164, 205, 246, 287, 328, 369, 410**

you can rebuild the answers by working with the digits **4 and 1**.

**Split → Multiply → Carry → Combine**

This idea can also be applied to other multi-digit numbers.

## Example: 93

Split **93** into its digits:

**93 → 9 | 3**

Now multiply each digit by the same number from **1 to 10**, handling the carry from right to left.

| | 9 | 3 | Result |
|---|---|---|---|
| ×1 | 9 × 1 = 9 | 3 × 1 = 3 | **93** |
| ×2 | 9 × 2 = 18 | 3 × 2 = 6 | **186** |
| ×3 | 9 × 3 = 27 | 3 × 3 = 9 | **279** |
| ×4 | 9 × 4 = 36 | 3 × 4 = 12 | **372** |
| ×5 | 9 × 5 = 45 | 3 × 5 = 15 | **465** |
| ×6 | 9 × 6 = 54 | 3 × 6 = 18 | **558** |
| ×7 | 9 × 7 = 63 | 3 × 7 = 21 | **651** |
| ×8 | 9 × 8 = 72 | 3 × 8 = 24 | **744** |
| ×9 | 9 × 9 = 81 | 3 × 9 = 27 | **837** |
| ×10 | 9 × 10 = 90 | 3 × 10 = 30 | **930** |

For example, **93 × 7**:

**9 × 7 = 63**
**3 × 7 = 21**

Start from the right:

- Write **1**
- Carry **2**
- Add the carry to 63: **63 + 2 = 65**

So:

**93 × 7 = 651**

The trick helps you **rebuild the table instead of memorizing every answer individually**.

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# Multiply TRICKS

## Multiply by 5

Multiply by 10, then divide by 2.

**68 × 5**

= 680 ÷ 2
= **340**

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## Multiply by 9

Multiply by 10, then subtract the number.

**47 × 9**

= 470 − 47
= **423**

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## Multiply by 11

For a two-digit number, put the sum of the digits between them.

**43 × 11**

4 + 3 = 7

**Answer: 473**

**68 × 11**

6 + 8 = 14

Carry the 1:

**Answer: 748**

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## Multiply by 25

Divide by 4, then multiply by 100.

**48 × 25**

= 48 ÷ 4 × 100
= 12 × 100
= **1200**

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### Multiply by 50

Divide by 2, then multiply by 100.

**76 × 50**

= 76 ÷ 2 × 100
= 38 × 100
= **3800**

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### Multiply by 125

Divide by 8, then multiply by 1000.

**32 × 125**

= 32 ÷ 8 × 1000
= 4 × 1000
= **4000**

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# PERCENTAGE & FRACTION TRICKS

### Find Percentages Using 10%

First find **10%** of the number. Then use it to build other percentages.

**Example: Find 15% of 480**

10% of 480 = **48**

5% is half of 10%:
48 ÷ 2 = **24**

So:

15% = 10% + 5%
= 48 + 24
= **72**

**Quick idea:**
Once you know **10%**, you can easily find **5%, 20%, 15%, 25%**, and more.

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## Common Fraction Shortcuts

50% = 1/2
25% = 1/4
20% = 1/5
12.5% = 1/8
10% = 1/10
5% = 1/20
2.5% = 1/40

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## 25% Means One Quarter

**25% of 480**

= 480 ÷ 4
= **120**

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## 12.5% Means One Eighth

**12.5% of 800**

= 800 ÷ 8
= **100**

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## Find 1% First

**7% of 350**

1% = 3.5

7% = 3.5 × 7

**Answer: 24.5**

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## Percentage of a Percentage

**20% of 15% of 500**

15% of 500 = 75

20% of 75 = **15**

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### Find the Original Value

Sometimes you know that a number represents a certain percentage of the original amount, and you need to find the original.

**Example:**

A price after a **15% increase** is **₹920**. What was the original price?

After a 15% increase, the new price is:

**100% + 15% = 115%**

So ₹920 represents **115%** of the original.

To find 100%:

**₹920 ÷ 1.15 = ₹800**

So the **original price was ₹800**.

**Quick idea:**
If the final value represents **115%**, divide it by **1.15** to get the original 100%.

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### Successive Percentage Changes

Increase by 20%:

100 → 120

Decrease by 20%:

120 → 96

So the overall change is:

**4% decrease**

The second 20% is calculated from 120, not 100.

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# PROFIT, LOSS & INTEREST TRICKS

### Profit Percentage

**Profit % = Profit ÷ Cost Price × 100**

Cost = ₹500
Selling = ₹600

Profit = ₹100

100 ÷ 500 × 100 = **20%**

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## Direct Selling Price

Cost = ₹800
Profit = 15%

Selling price:

800 × 1.15 = **₹920**

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## Reverse Cost Price

Selling price = ₹920
Profit = 15%

₹920 represents 115%.

920 ÷ 1.15 = **₹800**

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## Discount Shortcut

A 15% discount means you pay:

**85%**

₹2000 × 0.85 = **₹1700**

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## Successive Discounts

10% followed by 20%:

100 → 90 → 72

Total discount:

**28%**

Not 30%.

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## Simple Interest

**Simple Interest** is the extra money earned or paid on the original amount over time.

**Formula:**

**SI = (P × R × T) ÷ 100**

Where:

- **P** = Principal, the original amount
- **R** = Interest rate per year
- **T** = Time in years

**Example:**

You invest **₹5,000** at **8% per year** for **2 years**.

First find 8% of ₹5,000:

**8% of ₹5,000 = ₹400**

For 2 years:

**₹400 × 2 = ₹800**

So, the **Simple Interest = ₹800**.

**Total amount = ₹5,000 + ₹800 = ₹5,800**

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## Compound Growth

With compound growth, each year's increase is added to the amount. The next year's percentage is calculated on the **new amount**, not the original amount.

**Example: ₹1,000 grows by 10% each year for 2 years**

**Year 1:**
10% of ₹1,000 = ₹100
₹1,000 + ₹100 = **₹1,100**

**Year 2:**
10% of ₹1,100 = ₹110
₹1,100 + ₹110 = **₹1,210**

**Final amount: ₹1,210**

**Quick idea:**
For compound growth, **each new amount becomes the base for the next increase.**

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## Two-Year Compound Interest Shortcut

For **2 years**, you can quickly find the difference between **Compound Interest (CI)** and **Simple Interest (SI)** using:

**CI − SI = P × (R/100)²**

Where:

- **P** = original amount
- **R** = yearly interest rate

**Example: ₹10,000 at 10% per year for 2 years**

First, convert 10% to decimal:

**10 ÷ 100 = 0.1**

Then:

**₹10,000 × 0.1²**
**=** ₹10,000 × 0.01
**= ₹100**

So, **CI is ₹100 more than SI** after 2 years.

**Quick idea:**
For a **2-year** calculation, this shortcut gives you the CI–SI difference directly, without calculating both separately.

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## Remember the Base

Always ask:

**“The percentage is calculated on what?”**

Profit and loss percentages normally use **cost price** as the base.

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# DIVISIBILITY TRICKS

## Divisible by 2

Check the last digit.

**0, 2, 4, 6, 8 → YES**

Example:

48,732 → **Yes**

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## Divisible by 3

Add the digits.

57,624:

5 + 7 + 6 + 2 + 4 = 24

24 is divisible by 3.

**Answer: Yes**

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## Divisible by 4

Check the last two digits.

2,316 → last two digits = 16

16 is divisible by 4.

**Answer: Yes**

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## Divisible by 5

Last digit must be:

**0 or 5**

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## Divisible by 6

Must be divisible by both:

**2 and 3**

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## Divisible by 8

Check the last three digits.

12,344 → 344 ÷ 8 = 43

**Answer: Yes**

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## Divisible by 9

Add the digits.

73,152:

7 + 3 + 1 + 5 + 2 = 18

18 is divisible by 9.

**Answer: Yes**

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## Divisible by 10

Last digit must be:

**0**

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## Divisible by 11

Find the difference between alternating digit sums.

48,273:

(4 + 2 + 3) − (8 + 7)

= 9 − 15
= −6

6 is not divisible by 11.

**Answer: No**

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## Divisible by 12

Check divisibility by both:

**3 and 4**

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## Divisible by 15

Check divisibility by both:

**3 and 5**

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## Divisible by 25

Last two digits must be:

**00, 25, 50, or 75**

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## Digital Root: A Quick Calculation Check

**What is it?**
A digital root is the single digit you get after repeatedly adding the digits of a number. It can be used to quickly check whether a calculation is likely correct.

**Example:**

7583
→ 7 + 5 + 8 + 3 = 23
→ 2 + 3 = 5

So, the digital root of 7583 is 5.

**Quick use:**
Use digital roots as a fast error check when doing calculations. If the digital roots don't match, your answer is definitely wrong. If they match, the answer may be correct, but it is not proof.

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### Last Digit of Powers

When a number is multiplied by itself, its last digit often follows a repeating pattern.

**For example, powers of 2:**

- $2^1 = 2 \rightarrow$ last digit 2
- $2^2 = 4 \rightarrow$ last digit 4
- $2^3 = 8 \rightarrow$ last digit 8
- $2^4 = 16 \rightarrow$ last digit 6
- $2^5 = 32 \rightarrow$ last digit 2 again

So the last digits repeat:

$2 \rightarrow 4 \rightarrow 8 \rightarrow 6 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 6...$

The pattern has 4 numbers.

**Example:** Find the last digit of $2^{17}$

Since the pattern repeats every 4 powers:

17 ÷ 4 = 4 remainder 1

A remainder of 1 means we use the 1st number in the pattern:

2

So the last digit of $2^{17}$ is 2.

**Why is this useful?**
You can find the last digit of very large powers without calculating the entire number.

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# FAST CALCULATION TRICKS

### Round & Correct

When a number is close to a round number, calculate with the round number first, then correct the difference.

**998 + 347**

1000 + 347 = 1347
998 is 2 less than 1000:

1347 − 2 = **1345**

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## Group Numbers

Look for numbers that make **10, 100, 1000**, etc.

**27 + 38 + 73 + 62**

(27 + 73) + (38 + 62)
= 100 + 100
= **200**

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## Subtract by Compensation

Round the number you're subtracting, then add back the extra amount.

**1000 − 487**

1000 − 500 = 500

500 − 487 = 13, so add it back:

500 + 13 = **513**

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## Subtract From 1000

For numbers below 1000, subtract each digit from **9**, except the last digit, which is subtracted from **10**.

**1000 − 327**

9 − 3 = 6
9 − 2 = 7
10 − 7 = 3

**Answer: 673**

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## Double & Halve

You can **double one number and halve the other** without changing the answer.

**25 × 48**

25 × 48
= 50 × 24
= **1200**

This works because doubling one factor and halving the other keeps the product unchanged.

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## Multiply Near 100

When both numbers are close to 100, use their deficits from 100.

**98 × 97**

Deficits: 2 and 3

98 − 3 = **95**

2 × 3 = **06**

Put them together:

**9506**

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## Multiply by 101

Multiplying by 101 is the same as multiplying by 100 and adding the original number.

**47 × 101**

47 × 100 = 4700
4700 + 47 = **4747**

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## Divide by 5

To divide by 5 quickly, **double the number and divide by 10**.

**360 ÷ 5**

360 × 2 = 720
720 ÷ 10 = **72**

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## Divide by 25

Since **25 × 4 = 100**, multiply by 4 and divide by 100.

**700 ÷ 25**

700 × 4 = 2800
2800 ÷ 100 = **28**

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### Estimate Before You Calculate

Before doing a calculation, make a quick estimate. It helps you spot answers that are obviously too large or too small.

**398 × 21**

Approximate:

400 × 20 = **8000**

Exact:

398 × 21 = **8358**

The exact answer is reasonably close to the estimate, so it passes a quick sanity check.

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# MATH MAGIC TRICKS

## Predict the Total

You can know the final total **before anyone finishes adding**.

### The Trick

Ask someone to write any **5-digit number**.

**Their number:**
47,382

Now create your number by subtracting each digit from **9**:

9 − 4 = 5
9 − 7 = 2
9 − 3 = 6
9 − 8 = 1
9 − 2 = 7

**Your number:**
52,617

These two numbers always add to:

**47,382 + 52,617 = 99,999**

Now let them write another 5-digit number.

**63,915**

Again, create its partner:

**36,084**

So:

**63,915 + 36,084 = 99,999**

## The Prediction

The first four numbers always total:

**99,999 + 99,999 = 199,998**

Now let them write one final 5-digit number.

Suppose they write:

**12,345**

The final total must be:

**199,998 + 12,345 = 212,343**

So you can write **212,343** as your prediction before the final addition is done.

## The Secret

Your number is always made so that each pair adds to **99,999**.

**Quick formula:**

**Final Total = 199,998 + Final Number**